Resilient food solutions for Argentina under volcanic abrupt sunlight reduction scenarios: a linear optimization approach

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ASRS — volcanic, nuclear & impact scenarios

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Background

Motivation

Large-scale volcanic eruptions inject sulfate aerosols into the stratosphere, reducing incoming solar radiation for months to years — an Abrupt Sunlight Reduction Scenario (ASRS). The 1815 Tambora eruption and subsequent "year without a summer" offers the clearest historical precedent of the resulting agricultural collapse and food crises across the Northern Hemisphere.

Modern climate models confirm that eruptions generating $5-150$ Tg of stratospheric particulate can reduce global caloric crop production by $7-90\%$, depending on magnitude. Because existing food stocks can sustain populations for only a few months, a rapid adaptive response is required. This study focuses on Argentina as a strategic case study for the Southern Hemisphere.

Research aim

Objective

To evaluate the effectiveness of a phased portfolio of adaptive food-system interventions on per-capita caloric availability in Argentina across a seven-year post-eruption window, modeled under volcanic ASRS conditions spanning 5 to 150 Tg of atmospheric particulate loading.

Specific aims:

1. Quantify production loss under volcanic ASRS without adaptation
2. Model phased interventions and their combined caloric yield
3. Assess regional food-security and export capacity implications

At a glance

Key results

70%	3-6×
Population fed without adaptation (150 Tg)	Min. requirement met with full adaptation
300M	7 yr
People in the region who could receive surplus exports	Modeling window post-event

All scenarios modeled against a 2,100 kcal/day/person minimum requirement. Under the most severe scenario (150 Tg), full intervention brings net production to 7,800-14,000 kcal/day/person.

Methods

Methodology

Model framework
Adapted from the ALLFED Integrated Food System Model (Rivers et al., 2023), implemented in Python. Linear optimization minimizes starvation while maximizing caloric availability across the population, distributing food stocks and new production monthly over 120 months.
Crop yield loss estimates drawn from Xia et al. (2022), covering six particulate loading levels (5–150 Tg). Baseline production data from FAO (2020–2022).

Scenarios modeled

Scenario	Description	Adapt.
1a / 1b	Pre-event baseline (net / gross)	none
2a	150 Tg, no adaptation	none
2b	Redirection from livestock & biofuels	partial
2c	+ resilient food solutions	moderate
2d	+ greenhouse scale-up & area expansion	full

Country selection via multi-criteria decision matrix: supply-chain resilience (WEF GCI 2020), governance effectiveness (World Bank WGI 2021), and ASRS food production capacity.

Case selection

Why Argentina?

Southern hemisphere location
Temperate zone; less exposed to volcanic winter cooling concentrated in the Northern Hemisphere, as Tambora (1815) demonstrated.

Agricultural capacity
154M ha total agricultural land; 43M ha easily cultivable. Pre-ASRS gross production ~28,000 kcal/day/person.

Diversified economy
Strong agri-industrial base; capacity to repurpose paper mills, biorefineries, and livestock infrastructure rapidly.

Regional connectivity
Rail links with 6 neighbours; maritime trade with Brazil (key fossil fuel and food exporter), enabling surplus distribution.

Uruguay and Chile were also identified as priority candidates in the regional prioritization matrix. Argentina selected for its higher production capacity under modeled scenarios.

Adaptive strategies

Interventions evaluated

Livestock & biofuel redirection
Scaling down cattle and biofuel production releases grain currently consumed by animals. Estimated release: up to ~15,000 kcal/capita/day. Single highest-impact intervention (+300% availability vs. scenario 2a).

Cold-tolerant crop relocation
Introducing northern-hemisphere cold-tolerant varieties (wheat, potato, oat, rapeseed) to latitudes where soy and maize yields collapse. Relocation should begin immediately after event.

Low-tech greenhouse scale-up
Rapid expansion from 6,500 ha to ~2.3M ha within 2 years (×360). Controls temperature variability under reduced insolation. Scenario 2d key driver.

Cultivated area expansion
From 31M ha (current 5 crops) to 43M ha (easily cultivable) or up to 108M ha. With expansion to 108M ha, gross production could sustain 6–22× national population.

Seaweed cultivation revival
Argentine seaweed industry nearly extinct since mid-20th century. High potential along Chubut coast. Knowledge transfer from countries with active algae industries required.

Rationing & waste reduction
Assumed in baseline model: equitable distribution across population; reduction of food loss due to scarcity. Necessary for all scenarios to prevent unequal access collapse.

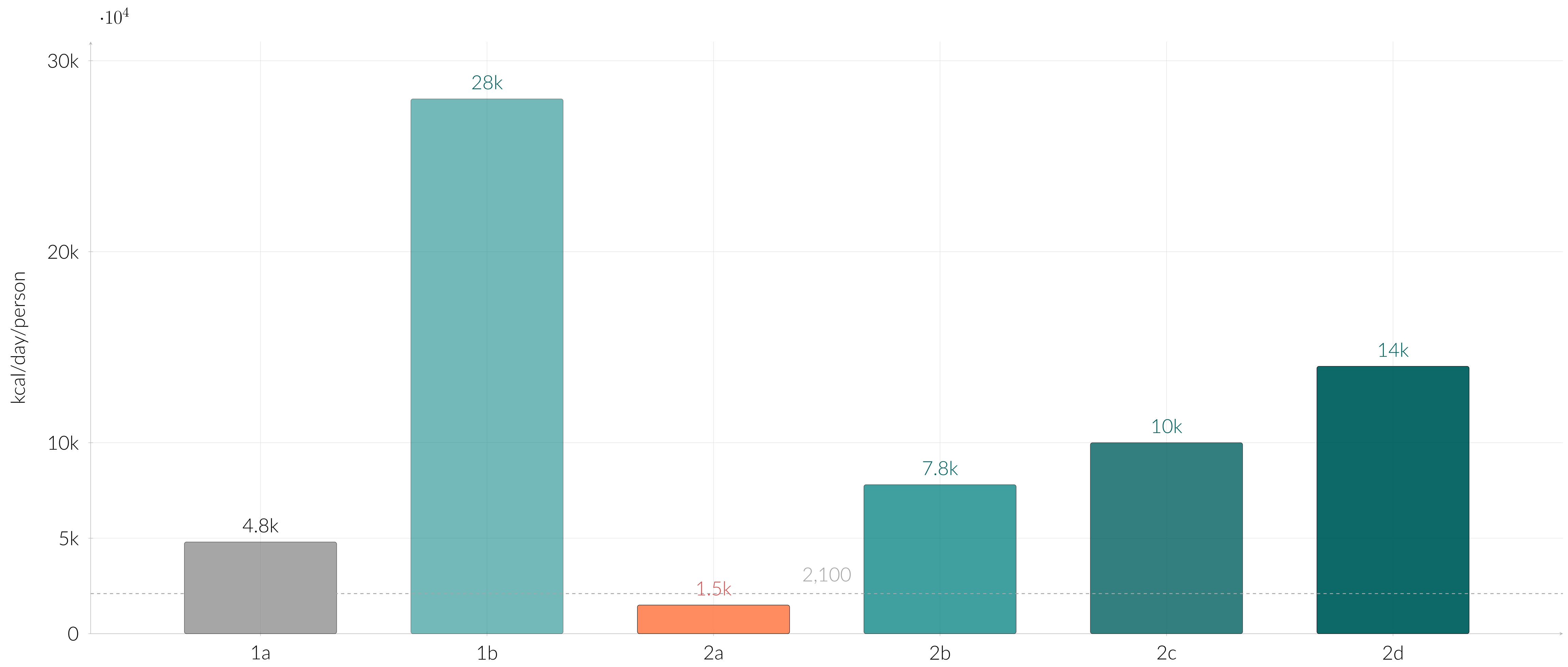
Industrial sugar (lignocellulosic)
Rapid repurposing of pulp and paper mills into lignocellulosic sugar factories. Contributes ~200 kcal/person/day in months 15–35 (scenario 2c/2d).

Single-cell protein (SCP)
Methane-based SCP production to compensate animal protein losses. Used as animal feed substitute in scenarios 2c/2d to free crop production for human consumption.

■ Supply redirection & agricultural adaptation ■ Industrial & high-technology solutions

Results

Caloric availability by scenario



Discussion

Conclusions

- A volcanic ASRS of 150 Tg would reduce Argentina's net food production to ~70% of minimum population requirements - famine without intervention, even with rationing alone.
- Redirecting food from livestock and biofuels - the most impactful single measure - can alone increase human-available calories by up to 300%, bringing production above the survival threshold.
- A phased combination of all modeled interventions ($2d$) produces $3-6\times$ minimum national requirements, enabling surplus export to support up to 300 million additional people in the region.
- Southern Cone countries occupy a structurally advantageous position in any ASRS scenario, including volcanic origin; advance national planning is both technically feasible and strategically urgent.
- Success depends on pre-event institutional preparedness: contingency plans, strategic input stockpiles, legislative flexibility, and inter-provincial coordination within Argentina's federal structure.

Caveats

Key uncertainties

Results are sensitive to assumptions that require further research:

- Volcanic aerosol optical depth
- Regional agroclimate response
- Algae scale-up feasibility
- Nutritional adequacy of resilient diets
- International trade continuity
- Federal coordination heterogeneity
- Demographic nutritional variation
- Industrial system disruption depth

Yield loss estimates approximate for cold-sensitive crops (maize, soy) vs. cold-tolerant crops (potato, wheat, oat). Local edaphological and hydrological variation across Argentine provinces not yet captured.

Agenda

Future work

Priority research directions to reduce uncertainty and increase policy relevance:

- Argentina-specific agroclimate models calibrated to volcanic forcing scenarios
- Province-level food distribution and economic modeling
- Pilot trials for low-tech greenhouse deployment and algae cultivation in Patagonia
- Subgroup-specific nutritional needs analysis under ASRS conditions
- Comparative analysis: volcanic vs. nuclear ASRS forcing on Southern Cone food systems

References & data

Selected references

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Model code: `https://github.com/allfed/allfed-integrated-model`
Supplementary material: `https://doi.org/10.5281/zenodo.11658966`
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